Model	Arch.	Layer	Acc.
S4	No gate	S4	18.3
-	No gate	S6	**97.0**
H3	H3	S4	57.0
Hyena	H3	Hyena	30.1
-	H3	S6	**99.7**
-	Mamba	S4	56.4
-	Mamba	Hyena	28.4
Mamba	Mamba	S6	**99.8**

Table 1: (**Selective Copying**.) Accuracy for combinations of architectures and inner sequence layers.

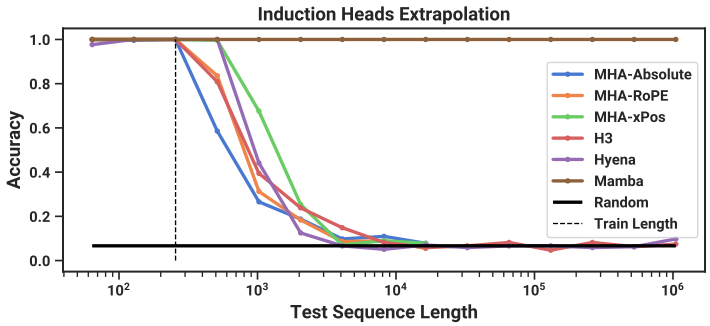


Table 2: (**Induction Heads**.) Models are trained on sequence length $2^8 = 256$, and tested on increasing sequence lengths of $2^6 = 64$ up to $2^{20} = 1048576$. Full numbers in Table 11.

4.1.2 Induction Heads

Induction heads (Olsson et al. 2022) is a simple task from the mechanistic interpretability lens (Elhage et al. 2021) that is surprisingly predictive of the in-context learning ability of LLMs. It requires models to perform associative recall and copy: for example, if the model has seen a bigram such as "Harry Potter" in the sequence, then the next time "Harry" appears in the same sequence, the model should be able to predict "Potter" by copying from history.

Dataset. We train a 2-layer model on the induction heads task at sequence length 256, with a vocab size of 16, which is comparable to prior work on this task (Dao, Fu, Saab, et al. 2023) but with longer sequences. We additionally investigate generalization and extrapolation abilities by evaluating on a range of sequence lengths from $2^6 = 64$ up to $2^{20} = 1048576$ at test time.

Models. Following established work on induction heads, we use 2 layer models, which allows attention to mechanistically solve the induction heads task (Olsson et al. 2022). We test both multi-head attention (8 heads, with various positional encodings) and SSM variants. We use a model dimension D of 64 for Mamba and 128 for the other models.

Results. Table 2 shows that Mamba—or more precisely, its selective SSM layer—has the ability to solve the task perfectly because of its ability to selectively remember the relevant token while ignoring everything else in between. It generalizes perfectly to million-length sequences, or 4000× longer than it saw during training, while no other method goes beyond 2×.

Out of positional encoding variants for attention models, xPos (which was designed for length extrapolation) is slightly better than the others; also note that all attention models were only tested up to sequence length $2^{14} = 16384$ due to memory limitations. Out of other SSMs, H3 and Hyena are similar, contrary to the findings in Poli et al. (2023).

4.2 Language Modeling

We evaluate the Mamba architecture on standard autoregressive language modeling against other architectures, on both pretraining metrics (perplexity) and zero-shot evaluations. We set the model sizes (depth and width) to mirror GPT3 specifications. We use the Pile dataset (L. Gao, Biderman, et al. 2020), and follow the training recipe described in Brown et al. (2020). All training details are in Appendix E.2.

4.2.1 Scaling Laws

For baselines, we compare against the standard Transformer architecture (GPT3 architecture), as well as the strongest Transformer recipe we know of (here referred to as Transformer++), based on the PaLM and LLaMa

Table 3: (**Zero-shot Evaluations**.) Best results for each size in bold. We compare against open source LMs with various tokenizers, trained for up to 300B tokens. Pile refers to the validation split, comparing only against models trained on the same dataset and tokenizer (GPT-NeoX-20B). For each model size, Mamba is best-in-class on every single evaluation result, and generally matches baselines at twice the model size.

Model	Token.	Pile ppl ↓	LAMBADA ppl ↓	LAMBADA acc ↑	HellaSwag acc ↑	PIQA acc ↑	Arc-E acc ↑	Arc-C acc ↑	WinoGrande acc ↑	Average acc ↑
Hybrid H3-130M	GPT2	—	89.48	25.77	31.7	64.2	44.4	24.2	50.6	40.1
Pythia-160M	NeoX	29.64	38.10	33.0	30.2	61.4	43.2	24.1	**51.9**	40.6
Mamba-130M	NeoX	**10.56**	**16.07**	**44.3**	**35.3**	**64.5**	**48.0**	**24.3**	**51.9**	**44.7**
Hybrid H3-360M	GPT2	—	12.58	48.0	41.5	68.1	51.4	24.7	54.1	48.0
Pythia-410M	NeoX	9.95	10.84	51.4	40.6	66.9	52.1	24.6	53.8	48.2
Mamba-370M	NeoX	**8.28**	**8.14**	**55.6**	**46.5**	**69.5**	**55.1**	**28.0**	**55.3**	**50.0**
Pythia-1B	NeoX	7.82	7.92	56.1	47.2	70.7	57.0	27.1	53.5	51.9
Mamba-790M	NeoX	**7.33**	**6.02**	**62.7**	**55.1**	**72.1**	**61.2**	**29.5**	**56.1**	**57.1**
GPT-Neo 1.3B	GPT2	—	7.50	57.2	48.9	71.1	56.2	25.9	54.9	52.4
Hybrid H3-1.3B	GPT2	—	11.25	49.6	52.6	71.3	59.2	28.1	56.9	53.0
OPT-1.3B	OPT	—	6.64	58.0	53.7	72.4	56.7	29.6	59.5	55.0
Pythia-1.4B	NeoX	7.51	6.08	61.7	52.1	71.0	60.5	28.5	57.2	55.2
RWKV-1.5B	NeoX	7.70	7.04	56.4	52.5	72.4	60.5	29.4	54.6	54.3
Mamba-1.4B	NeoX	**6.80**	**5.04**	**64.9**	**59.1**	**74.2**	**65.5**	**32.8**	**61.5**	**59.7**
GPT-Neo 2.7B	GPT2	—	5.63	62.2	55.8	72.1	61.1	30.2	57.6	56.5
Hybrid H3-2.7B	GPT2	—	7.92	55.7	59.7	73.3	65.6	32.3	61.4	58.0
OPT-2.7B	OPT	—	5.12	63.6	60.6	74.8	60.8	31.3	61.0	58.7
Pythia-2.8B	NeoX	6.73	5.04	64.7	59.3	74.0	64.1	32.9	59.7	59.1
RWKV-3B	NeoX	7.00	5.24	63.9	59.6	73.7	67.8	33.1	59.6	59.6
Mamba-2.8B	NeoX	**6.22**	**4.23**	**69.2**	**66.1**	**75.2**	**69.7**	**36.3**	**63.5**	**63.3**
GPT-J-6B	GPT2	–	4.10	68.3	66.3	75.4	67.0	36.6	64.1	63.0
OPT-6.7B	OPT	–	4.25	67.7	67.2	76.3	65.6	34.9	65.5	62.9
Pythia-6.9B	NeoX	6.51	4.45	67.1	64.0	75.2	67.3	35.5	61.3	61.7
RWKV-7.4B	NeoX	6.31	4.38	67.2	65.5	76.1	67.8	37.5	61.0	62.5

total of $2^{20} \approx 1M$ tokens per batch. Models were trained for $10K$ gradient steps for a total of $10B$ tokens.

Results. Figure 5 (Left) shows that Mamba's pretraining perplexity improves smoothly with model size, and that Mamba scales better than both HyenaDNA and Transformer++. For example, at the largest model size of $\approx 40M$ parameters, the curve shows that Mamba can match the Transformer++ and HyenaDNA models with roughly $3\times$ to $4\times$ fewer parameters.

4.3.2 Scaling: Context Length

In the next DNA experiment, we investigate the scaling properties of models with respect to sequence length. We only compare the HyenaDNA and Mamba models, as quadratic attention becomes prohibitively expensive at longer sequence lengths. We pretrain models on sequence lengths $2^{10} = 1024$, $2^{12} = 4096$, $2^{14} = 16384$, $2^{16} = 65536$, $2^{18} = 262144$, $2^{20} = 1048576$. We fix a model size of 6 layers by width 128 (about 1.3M-1.4M parameters). Models were trained for $20K$ gradient steps for a total of $\approx 330B$ tokens. The longer sequence lengths used sequence length warmup similar to (Nguyen, Poli, et al. 2023).

Results. Figure 5 (Right) shows that Mamba is able to make use of longer context even up to extremely long sequences of length 1M, and its pretraining perplexity improves as the context increases. On the other hand, the HyenaDNA model gets worse with sequence length. This is intuitive from the discussion in Section 3.5 on properties of the selection mechanism. In particular, LTI models cannot selectively ignore information; from a convolutional perspective, a very long convolution kernel is aggregating all information across a long sequence

16000 Hz Pretraining details largely follow the standard language modeling setup (Section 4.2). Figure 7 evaluates the effect of increasing training sequence lengths from $2^{13} = 8192$ to $2^{20} \approx 10^6$, while keeping computation fixed. (There are some slight edge cases to the way the data is curated, which may lead to kinks in the scaling curves. For example, only minute-long clips were available so the maximum sequence length is actually bounded by $60s \cdot 16000Hz = 960000$.)

Both Mamba and the SaShiMi (S4+MLP) baseline improve consistently with longer context lengths; Mamba is better throughout, and the gap widens at longer lengths. The main metric is bits per byte (BPB), which is a constant factor $\log(2)$ of the standard negative log-likelihood (NLL) loss for pretraining other modalities.

We note one important detail: this is the only experiment in this paper in which we switched from the real parameterization to complex (Section 3.6). We show additional ablations in Appendix E.4.

4.4.2 Autoregressive Speech Generation

SC09 is a benchmark speech generation dataset (Donahue, McAuley, and Puckette 2019; Warden 2018), consisting of 1-second clips sampled at 16000 Hz of the digits "zero" through "nine" with highly variable characteristics. We largely follow the autoregressive training setup and generation protocol of Goel et al. (2022).

Table 4 shows automated metrics of the Mamba-UNet model compared to a variety of baselines from Goel et al. (2022): WaveNet (Oord et al. 2016), SampleRNN (Mehri et al. 2017), WaveGAN (Donahue, McAuley, and Puckette 2019), DiffWave (Z. Kong et al. 2021), and SaShiMi. A small Mamba model outperforms the state-of-the-art (and much larger) GAN- and diffusion- based models. A larger model parameter-matched to the baselines further improves on fidelity metrics dramatically.

Table 5 takes the small Mamba model and investigates combinations of different architectures for the outer stages and center stage. It shows that Mamba is consistently better than S4+MLP in the outer blocks, and Mamba > S4+MLP > MHA+MLP in the center blocks.

Table 4: (**SC09**) Automated metrics for unconditional generation on a challenging dataset of fixed-length speech clips. (*Top to Bottom*) Autoregressive baselines, non-autoregressive baselines, Mamba, and dataset metrics.

Model	Params	NLL ↓	FID ↓	IS ↑	mIS ↑	AM ↓
SampleRNN	35.0M	2.042	8.96	1.71	3.02	1.76
WaveNet	4.2M	1.925	5.08	2.27	5.80	1.47
SaShiMi	5.8M	1.873	1.99	5.13	42.57	0.74
WaveGAN	19.1M	-	2.03	4.90	36.10	0.80
DiffWave	24.1M	-	1.92	5.26	51.21	0.68
+ SaShiMi	23.0M	-	1.42	5.94	69.17	0.59
Mamba	6.1M	**1.852**	0.94	6.26	88.54	0.52
Mamba	24.3M	1.860	**0.67**	**7.33**	**144.9**	**0.36**
Train	-	-	0.00	8.56	292.5	0.16
Test	-	-	0.02	8.33	257.6	0.19

Table 5: (**SC09 Model Ablations**) Models with 6M parameters. In SaShiMi's U-Net backbone, there are 8 center blocks operating on sequence length 1000, sandwiched on each side by 8 outer blocks on sequence length 4000, sandwiched by 8 outer blocks on sequence length 16000 (40 blocks total). The architecture of the 8 center blocks are ablated independently of the rest. Note that Transformers (MHA+MLP) were not tested in the more important outer blocks because of efficiency constraints.

Outer	Center	NLL ↓	FID ↓	IS ↑	mIS ↑	AM ↓
S4+MLP	MHA+MLP	1.859	1.45	5.06	47.03	0.70
S4+MLP	S4+MLP	1.867	1.43	5.42	53.54	0.65
S4+MLP	Mamba	1.859	1.42	5.71	56.51	0.64
Mamba	MHA+MLP	**1.850**	1.37	5.63	58.23	0.62
Mamba	S4+MLP	1.853	1.07	6.05	73.34	0.55
Mamba	Mamba	1.852	**0.94**	**6.26**	**88.54**	**0.52**

4.5 Speed and Memory Benchmarks

We benchmark the speed of the SSM scan operation (state expansion $N = 16$), as well as the end-to-end inference throughput of Mamba, in Figure 8. Our efficient SSM scan is faster than the best attention implementation that we know of (FlashAttention-2 (Dao 2023)) beyond sequence length 2K, and up to 20-40× faster than a standard scan implementation in PyTorch. Mamba achieves 4-5× higher inference throughput than a Transformer of similar size, since without the KV cache it can use much higher batch sizes. For example, a Mamba-6.9B (untrained) would have higher inference throughput than a 5× smaller Transformer-1.3B. Details in Appendix E.5, which additionally includes a benchmark of memory consumption.

Table 6: (**Ablations: Architecture and SSM layer**.) The Mamba block performs similarly to H3 while being simpler. In the inner layer, there is little difference among different parameterizations of LTI models, while selective SSMs (S6) provide a large improvement. More specifically, the S4 (real) variant is S4D-Real and the S4 (complex) variant is S4D-Lin.

Model	Arch.	SSM Layer	Perplexity
Hyena	H3	Hyena	10.24
H3	H3	S4 (complex)	10.30
-	H3	S4 (real)	10.34
-	H3	S6	**8.95**

Model	Arch.	SSM Layer	Perplexity
-	Mamba	Hyena	10.75
-	Mamba	S4 (complex)	10.54
-	Mamba	S4 (real)	10.56
Mamba	Mamba	S6	**8.69**

Table 7: (**Ablations: Selective parameters**.) Δ is the most important parameter (Theorem 1), but using multiple selective parameters together synergizes.

Selective Δ	Selective $\boldsymbol{B}$	Selective $\boldsymbol{C}$	Perplexity
✗	✗	✗	10.93
✗	✓	✗	10.15
✗	✗	✓	9.98
✓	✗	✗	9.81
✓	✓	✓	8.71

Table 8: (**Ablations: Parameterization of $\boldsymbol{A}$**.) The more standard initializations based on S4D-Lin (Gu, Gupta, et al. 2022) perform worse than S4D-Real or a random initialization, when the SSM is selective.

$\boldsymbol{A}_n$ Initialization	Field	Perplexity
$\boldsymbol{A}_n = -\frac{1}{2} + ni$	Complex	9.16
$\boldsymbol{A}_n = -1/2$	Real	8.85
$\boldsymbol{A}_n = -(n+1)$	Real	8.71
$\boldsymbol{A}_n \sim \exp(\mathcal{N}(0,1))$	Real	8.71

Table 9: (**Ablations: Expressivity of Δ**.) The selection mechanism of Δ constructs it with a projection of the input. Projecting it even to dim. 1 provides a large increase in performance; increasing it further provides further improvements at the cost of a modest increase in parameters. State size fixed to $N = 16$.

Size of Δ proj.	Params (M)	Perplexity
-	358.9	9.12
1	359.1	8.97
2	359.3	8.97
4	359.7	8.91
8	360.5	8.83
16	362.1	8.84
32	365.2	8.80
64	371.5	8.71

Table 10: (**Ablations: SSM state dimension**.) (*Top*) Constant $\boldsymbol{B}$ and $\boldsymbol{C}$ (*Bottom*) Selective $\boldsymbol{B}$ and $\boldsymbol{C}$. Increasing the SSM state dimension N, which can be viewed as an expansion factor on the dimension of the recurrent state, can significantly improve performance for a negligible cost in parameters/FLOPs, but only when $\boldsymbol{B}$ and $\boldsymbol{C}$ are also selective. Size of Δ projection fixed to 64.

State dimension N	Params (M)	Perplexity
1	367.1	9.88
2	367.4	9.86
4	368.0	9.82
8	369.1	9.82
16	371.5	9.81
1	367.1	9.73
2	367.4	9.40
4	368.0	9.09
8	369.1	8.84
16	371.5	8.71

5 Discussion

We discuss related work, limitations, and some future directions.

Related Work. Appendix A discusses how the selection mechanism relates to similar concepts. Appendix B has an extended related work of SSMs and other related models.

No Free Lunch: Continuous-Discrete Spectrum. Structured SSMs were originally defined as discretizations of continuous systems (1), and have had a strong inductive bias toward continuous-time data modalities such as perceptual signals (e.g. audio, video). As discussed in Sections 3.1 and 3.5, the selection mechanism overcomes their weaknesses on discrete modalities such as text and DNA; but this conversely can impede their performance